

Romeo Silvestri

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Summary

Applied Machine Learning Engineer with **3+ years of work experience** and a strong foundation in statistics and data science. Specializing in AI research, MLOps workflows, reinforcement learning and Generative AI, I design and deploy robust, scalable models. Author of peer-reviewed papers, eager to apply my technical expertise and research to deliver projects with real-world impact.

Skills

Languages & Frameworks: Python, R, SQL, FastAPI, Gymnasium, Numpy, Optuna, Pandas, PySpark, PyTorch, Scikit-learn, TensorFlow

Tools & Databases: Airflow, Docker, Git, LangChain, LangGraph, MLflow, Power BI, Prefect, Spark, MongoDB, MySQL, PostgreSQL

Techniques: A/B Testing, Agentic AI, Classification, Clustering, Deep Learning, Generative AI, Machine Learning, Model Monitoring, Natural Language Processing (NLP), Neural Networks, RAG, Reinforcement Learning, Regression, Time Series Forecasting

Work Experience

ML Engineer | Fondazione Bruno Kessler (FBK), Trento 12/2023 - Present

- Optimized ML pipelines and provided APIs in a research and production environment using **Python**, **FastAPI**, **GitLab**, **LangGraph**, and **MLflow**, while collaborating closely with domain experts to address complex problems and streamline processes.
- Designed, tested, and deployed a Decision Support System (DSS) for irrigation, integrating heterogeneous data sources, including IoT sensors and weather data, reducing operational costs by around **50%** across several companies.
- Developed a Deep Reinforcement Learning agent with PPO optimization, leveraging a digital twin environment using a surrogate and a physical model for simulation and policy training, with synthetic weather data generation via KNN bootstrap.
- Developed a multi-agent LLM-based chatbot for the European AI Factories project, leveraging pre-trained models and Retrieval-Augmented Generation (RAG) to support decision-making. The system provides real-time, data-driven advice on agriculture, integrating legal guidelines and best technical practices, with support for global and region-specific recommendation models.
- Built state-of-the-art time series forecasting models for sensor trends using ARIMA, XGBoost, and LSTM neural networks, achieving around **5%** prediction error through advanced feature engineering, hyperparameter tuning, and rigorous validation.

Data Scientist | Youbidata, Bologna 10/2023 - 11/2023

- Developed a centralized financial monitoring dashboard using **PostgreSQL** and **Power BI** to provide KPIs for a consulting client.

Statistician (Internship) | ISTAT, Rome 03/2023 - 08/2023

- Analyzed national demographic data using **Excel**, **R** and **SQL** to uncover socio-economic trends and produce periodic reports.
- Evaluated the Austrian quality framework's suitability for the Italian Census, achieving statistically significant results.

Education

M.Sc. in Statistics & Data Science | Sapienza University of Rome 11/2020 - 03/2023

Thesis: *Analyzing Forecasting Models for Real Estate Markets with a Case Study of Madrid Prices* Graduation Grade: 107/110

B.Sc. in Statistics | University of Padua 10/2017 - 11/2020

Thesis: *The Impact of Migration on Population Dynamics in Italian Provinces* Graduation Grade: 102/110

Publications

<i>A Calibrated Wasserstein Drift Detector for Residual-Based Monitoring of Regression Models</i>	Neurocomputing (under review)
<i>A Retrieval-Augmented LLM Framework for PRISMA-Aligned Systematic Reviews</i>	NCAA (under review)
<i>Deep Reinforcement Learning for Irrigation Optimization Based on Crop-Soil Dynamics</i>	IEEE MetroInd 2026
<i>Smart Irrigation with Fuzzy Decision Support Systems in Trentino Vineyards</i>	MDPI Sensors
<i>A Fuzzy Decision Support System to Optimize Irrigation Practices in Trentino Region</i>	IEEE CoDIT 2025
<i>Comparative Analysis of Soil Moisture Interpolation Techniques in Apple Orchards of Trentino Region</i>	IEEE MetroAgriFor 2024

Additional Information

Certifications: Deep Learning (DeepLearning.AI), Google Data Analytics, IBM Data Science

Teaching: Private Tutor in Statistics and IT (2020-2021), University Tutor at Sapienza University (2022)

Organizations: Italian Volunteers Blood Donation (AVIS), Mensa High IQ Society

Interests: Art, Chess, Mixed Martial Arts, Music, Poetry, Technology, Traveling

I authorize the processing of personal data contained in my curriculum vitae based on art. 13 GDPR 679/16.